

COLORADO PLAINS, CO, USA

WMO: 724698

Lat: 40.167N

Lon: 103.217W

Elev: 1421

StdP: 85.38

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24015

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -18.4	(c) -15.5	(d) -22.3	(e) 0.6	(f) -17.3	(g) -19.1	(h) 0.8	(i) -12.3	(j) 16.0	(k) -2.1	(l) 14.4	(m) -1.4	(n) 5.0	(o) 260	(p) 0.691

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	35.3	17.2	33.7	17.2	32.3	17.2	20.2	28.6	19.5	28.3	18.8	27.7	5.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.3	22.4	17.2	14.7	21.9	16.3	13.8	21.4	64.6	28.2	61.8	28.0	59.3	27.6	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	11.9	10.4	-23.1	38.0	3.0	1.4	-25.2	39.0	-27.0	39.9	-28.7	40.7	-30.8	41.7		
			-23.4	21.8	2.8	1.1	-25.4	22.6	-27.1	23.2	-28.6	23.9	-30.7	24.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-1.2	-0.4	4.8	8.5	13.8	20.1	23.8	22.6	18.0	10.3	4.0	-1.3		
	DBStd	10.39	6.15	6.54	6.07	5.46	4.99	4.18	3.13	3.07	4.90	5.72	6.10	6.18		
	HDD10.0	1545	348	293	178	89	21	0	0	6	65	193	351			
	HDD18.3	3393	606	524	420	296	153	29	3	6	64	252	431	608		
	CDD10.0	1656	1	2	17	44	141	302	428	390	244	74	12	1		
	CDD18.3	462	0	0	0	1	14	81	173	137	53	3	0	0		
	CDH23.3	5652	0	0	8	43	263	1060	2003	1481	695	95	3	0		
	CDH26.7	2584	0	0	0	4	79	473	1036	688	286	18	0	0		
(14) Wind	WSAvg	5.3	5.2	5.5	5.8	6.2	5.5	5.3	4.9	4.9	5.0	5.3	5.2	5.3		
(15) Precipitation	PrecAvg	399	8	6	24	33	76	65	71	45	25	20	12	7		
	PrecMax	586	30	22	80	88	172	182	147	133	84	84	36	22		
	PrecMin	268	0	0	2	7	8	16	11	3	0	2	0	0		
	PrecStd	85	7	5	21	22	45	41	30	37	23	20	10	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.5	19.7	24.4	27.1	32.1	37.0	37.5	36.2	34.5	28.9	22.9	16.5		
		MCWB	6.1	7.2	9.4	11.9	15.0	16.4	17.8	16.9	16.3	13.1	9.2	6.3		
	2%	DB	12.9	15.8	21.3	24.3	28.8	33.9	35.9	34.1	32.3	26.2	19.5	12.8		
		MCWB	4.4	5.8	8.4	10.8	14.3	16.7	17.5	17.1	15.6	11.9	8.1	4.5		
	5%	DB	10.0	12.3	18.3	22.0	26.4	31.8	34.1	32.6	30.2	23.2	16.4	10.1		
		MCWB	3.2	4.3	7.4	10.2	14.1	16.5	17.7	17.6	15.1	11.1	6.9	3.1		
	10%	DB	7.3	9.0	15.4	18.8	23.7	29.3	32.5	31.0	27.7	20.0	13.3	7.4		
		MCWB	1.8	2.6	6.1	9.2	13.3	16.4	17.7	17.6	15.0	9.9	5.3	1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	8.0	10.7	13.6	17.8	20.3	21.2	21.4	19.0	15.1	10.2	6.7		
		MCDB	15.1	18.1	20.9	23.1	26.5	28.9	29.5	29.3	28.6	24.8	20.1	15.8		
	2%	WB	4.9	6.1	9.1	11.9	16.3	19.1	20.3	20.2	17.7	13.2	8.7	4.9		
		MCDB	11.9	15.1	19.1	21.1	24.1	27.7	28.5	28.5	26.8	22.2	18.7	12.4		
	5%	WB	3.4	4.5	7.9	10.7	15.2	18.2	19.7	19.4	16.7	11.9	7.3	3.3		
		MCDB	9.5	11.7	17.4	19.9	23.4	26.9	28.5	27.9	25.3	20.3	16.0	9.4		
	10%	WB	1.9	2.9	6.5	9.6	14.1	17.4	19.0	18.6	15.8	10.7	5.7	1.8		
		MCDB	6.7	8.6	14.6	18.1	22.0	26.3	28.2	27.0	24.2	18.6	12.7	7.0		
(35) Mean Daily Temperature Range	MDBR		12.3	12.5	14.3	14.0	13.8	14.9	14.8	14.5	14.7	14.0	13.2	12.2		
	5% DB	MCDBR	15.8	17.0	18.5	18.6	17.7	18.1	17.2	16.9	17.8	18.3	17.2	15.5		
		MCWBR	9.2	9.4	9.0	8.6	6.9	6.1	5.1	5.2	6.0	7.9	8.7	9.0		
	5% WB	MCDBR	14.8	16.1	17.6	17.1	15.8	15.5	14.9	14.6	15.6	16.2	16.6	14.9		
		MCWBR	8.9	9.3	9.0	8.6	7.3	6.6	5.5	5.8	6.0	7.7	8.7	8.9		
(40) Clear-Sky Solar Irradiance	taub		0.242	0.251	0.272	0.297	0.323	0.333	0.359	0.348	0.311	0.275	0.253	0.243		
	taud		2.595	2.577	2.526	2.469	2.459	2.455	2.421	2.445	2.522	2.597	2.581	2.572		
	Ebn at Noon		957	990	992	979	953	939	911	915	935	946	932	927		
	Edh at Noon		67	79	93	106	110	111	114	108	93	77	67	63		
(44) All-Sky Solar Radiation	RadAvg		2.50	3.30	4.59	5.63	6.48	7.46	7.33	6.47	5.29	3.78	2.70	2.14		
	RadStd		0.14	0.23	0.34	0.34	0.47	0.33	0.28	0.21	0.25	0.28	0.17	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page